

Experiment: Design 1

Experiment Name: Differentiator and Integrator Op-Amp Circuit Design

Objective

The objective of this experiment was to analyze op-amp circuits in differentiator and integrator configurations and assess their relationship with Laplace Transforms.

Equipment

- (1) Computer running PSPICE software

*Note: This experiment was exclusively performed using simulation software.

Procedure

Part 1 – Differentiator Op-Amp Circuits

1. Open the PSICE software by searching “Capture” in the Windows menu and open a new file
2. Construct the circuit of figure 1 below by dropping parts, connecting with wires, and assigning values:

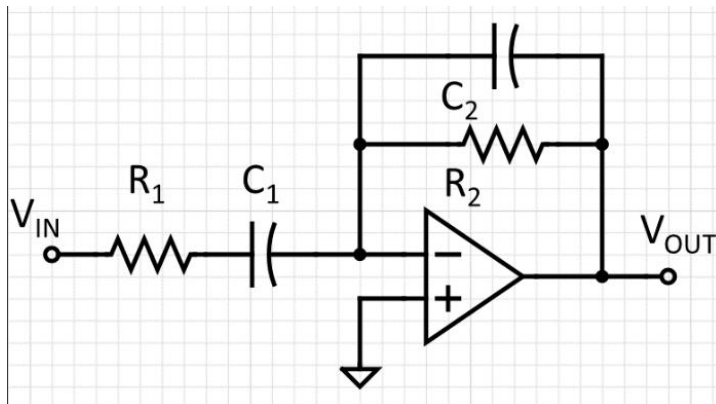


Figure 1: The following is the basis of our differentiator circuit with which we will change values to observe different responses in the time and frequency domain.

3. Using the circuit of figure 1 above, assign the following values:

V_{in} = “Vsin” source with an amplitude of 5V, and frequency of 1k Hz

$R_1 = 220\ \Omega$

$C_1 = 0.1\ \mu\text{F}$

$C_2 = 0.01\ \mu\text{F}$

$$R2 = 2.2k \Omega$$

*Note: the operational amplifier used is labeled “LM385” with positive and negative 15 V across the Vcc rails.

4. Make a new simulation profile and place differential voltage markers across the Vout as well as Vin.
5. Run the simulation, analyze and record results.
6. Reconstruct the circuit of figure 1 with the following changes:

Vin = “Vpulse” source with :

$$V1 = 5$$

$$V2 = 0$$

$$TD = 0$$

$$TR = 0.25 \text{ m}$$

$$TF = 0.25 \text{ m}$$

$$PW = 0.000001$$

$$PER = 0.5\text{m}$$

7. Repeat steps 4 and 5.
8. Reconstruct the circuit above with the following changes:
Vin = “Vac” with a voltage of 5 V
9. Make a new simulation profile, but adjust the characteristics of the waveform to run an AC sweep from 1 to 1k Hz with 100 points
10. Repeat steps 4 and 5.

Part 2 – Integrator Operational Amplifier Circuits

1. Using the same circuit construction techniques above reconstruct the circuit of figure 1 with the following modifications:

$V_{in} = "V_{sin}"$ with an amplitude of 5V, and frequency of 1k Hz

C1 is removed

$R1 = 5k\ \Omega$

$C2 = 0.125\ \mu F$

$R2 = 100k\ \Omega$

2. Make a new simulation profile reverting back to the standard time domain.
3. Place differential voltage markers as before.
4. Run the simulation and analyze the results.
5. Reconstruct the circuit constructed in step 1 with the following modification:

$V_{in} = "V_{pulse}"$ ranging in step voltage from 5 to -5 V

6. Repeat steps 2 – 4.
7. Reconstruct the circuit constructed in step 1 with the following modification:

$V_{in} = "V_{ac}"$ with values 1 to 0 V

8. Make a new simulation profile, but adjust the characteristics of the waveform to run an AC sweep from 1 to 1k Hz with 100 points
9. Record and analyze the circuit.

Results

Part 1 – Results

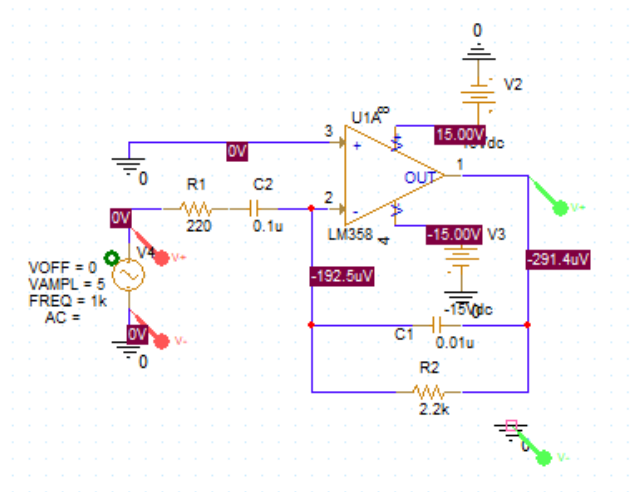


Figure 2: The following is differentiator configuration of figure 1 with a V_{sin} source voltage.

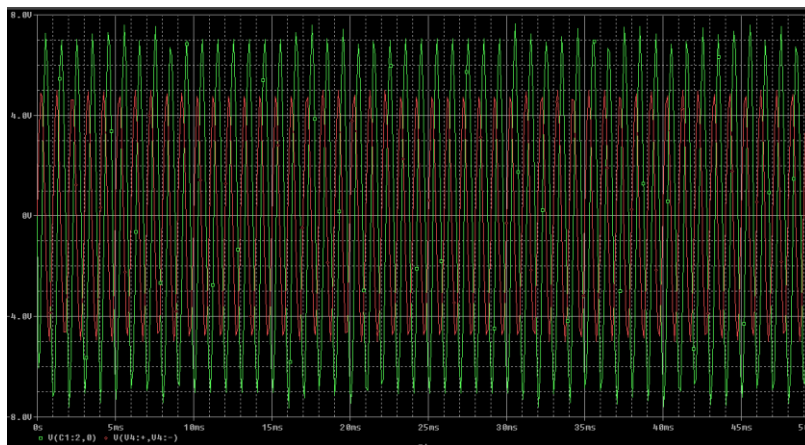


Figure 3: The following is the waveform of figure 2.

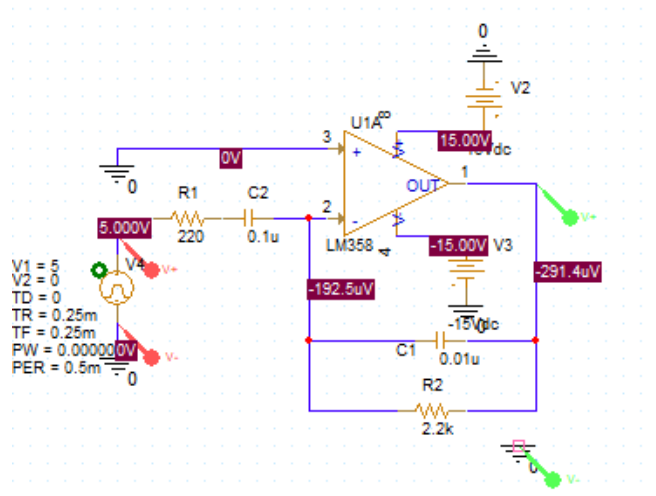


Figure 4: The following is differentiator configuration of figure 1 with a V_{pulse} source voltage.

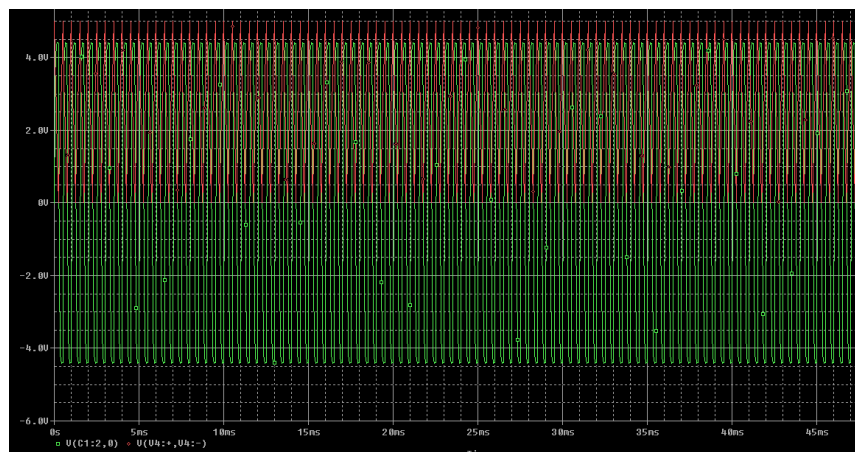


Figure 5: The following is the waveform of figure 4.

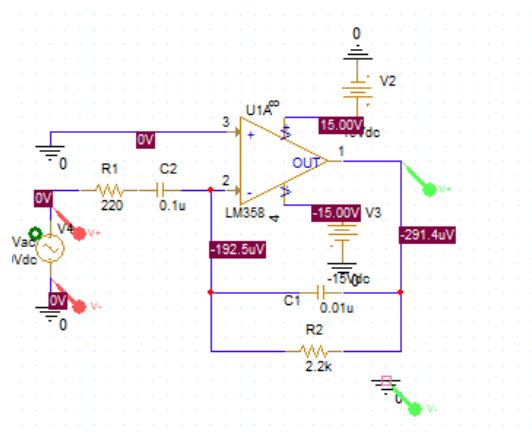


Figure 6: The following is differentiator configuration of figure 1 with an AC Sweep of Vac source voltage.

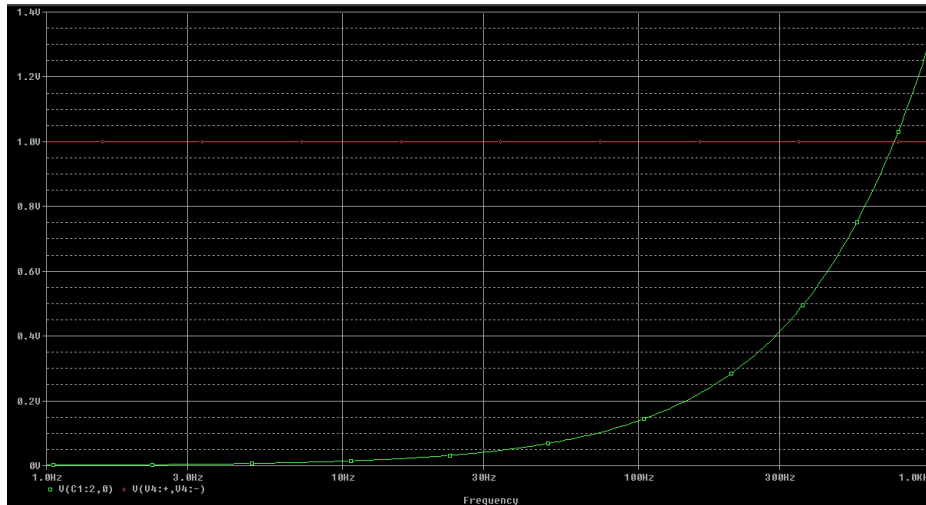


Figure 7: The following is the waveform of figure 6.

Part 2 – Results

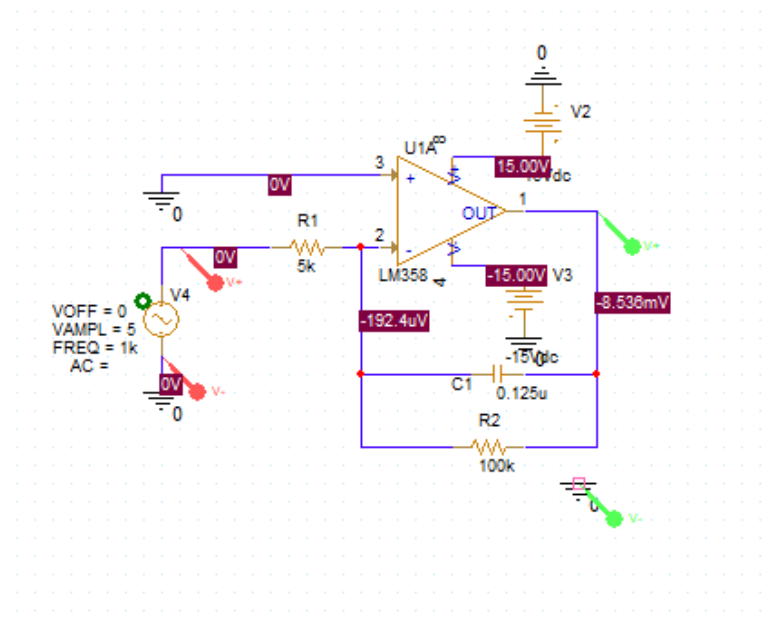


Figure 8: The following is integrator configuration of figure 1 with a Vsin source voltage.

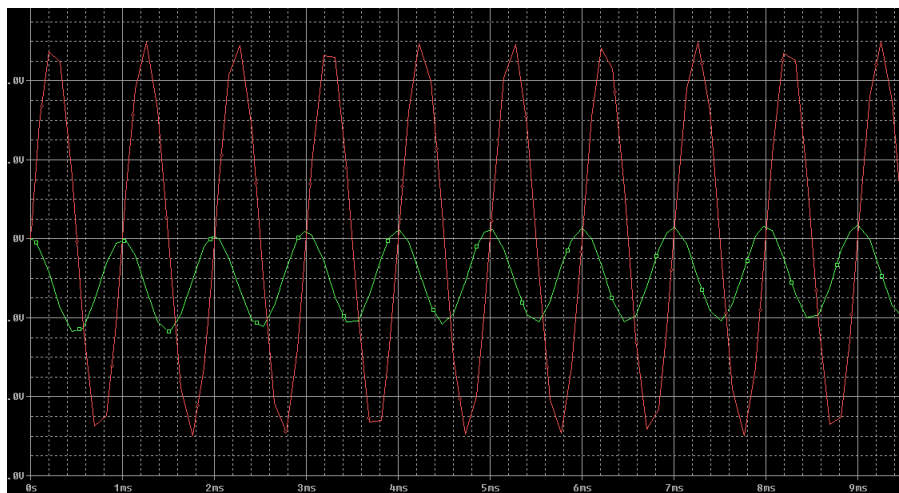


Figure 9: The following is the waveform of figure 8.

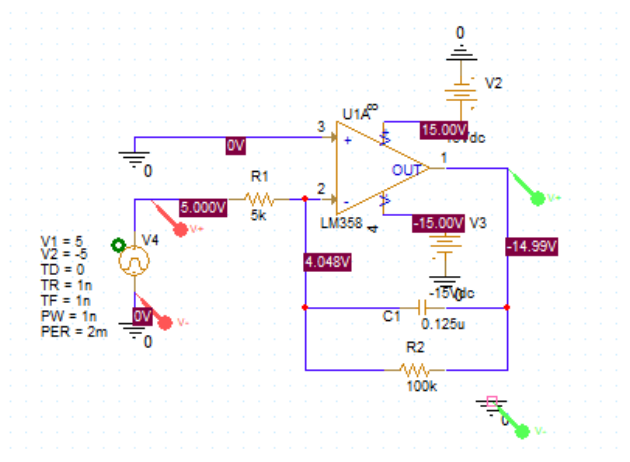


Figure 10: The following is integrator configuration of figure 1 with a V_{pulse} source voltage.

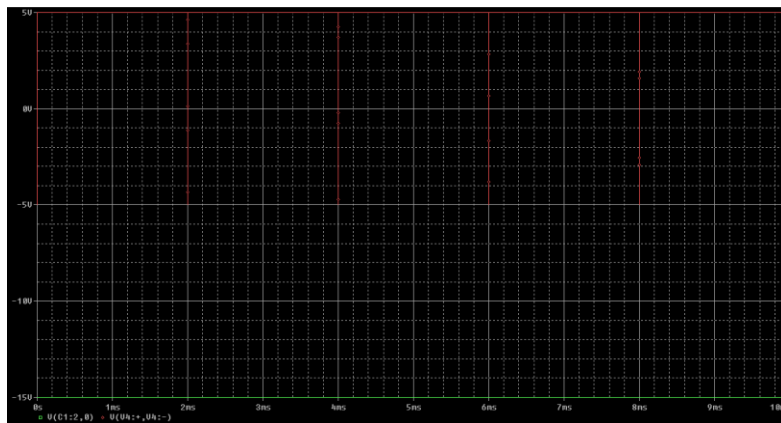


Figure 11: The following is the waveform of figure 10.

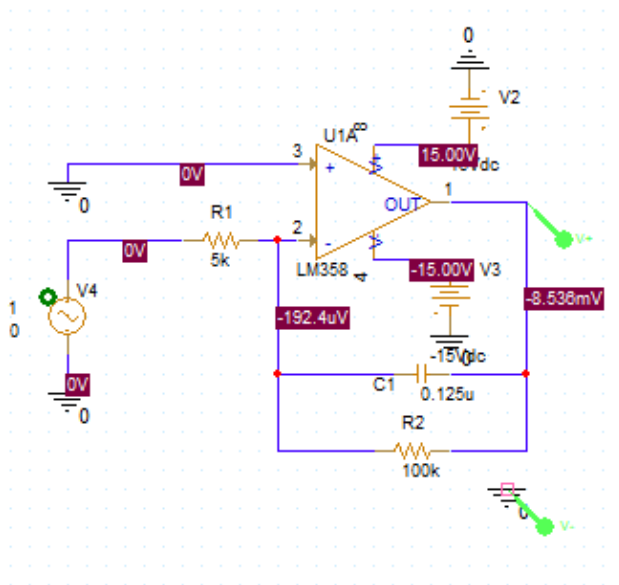


Figure 12: The following is integrator configuration of figure 1 with an AC Sweep of Vac source voltage.

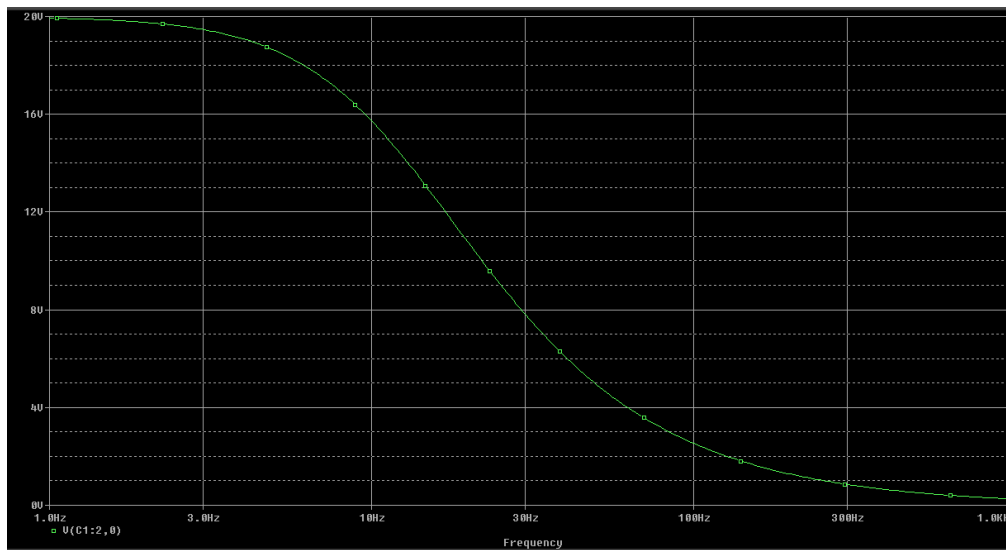


Figure 13: The following is the waveform of figure 12

Post Lab

No post lab problems were assigned for this experiment.

Conclusion

The general response we observed from the circuits is that a differentiator op-amp differentiates the input voltage at its output and an integrator does the opposite. For example, when feeding a V_{in} of a sin wave into the differentiator, we obtained a positive cosin function in the waveform output. Plugging that same source voltage into the integrator obtained a negative cosin. We observed the same functionality with the other source voltages in the time domain. Differentiating and integrating the AC sweep sources gave us exponential and logarithmic waveforms respectively which is what we would expect in the frequency domain.

What stood out to us the most was that the same operational amplifier could be configured to be either a differentiator or integrator with the correct supplementary circuit elements. Also, we should note the difficulties we had in producing the results as our computer did not find the PATH for the parts library which required us to change computers.

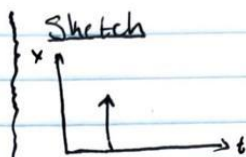
Preparation Problems

As the results were found by hand, all results may be seen by referencing the **Worksheet** section of this report. All answers are clearly labeled.

Worksheet

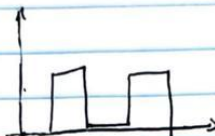
Pre Lab

1.) Function
impulse



$$\delta(t) \leftrightarrow 1$$

step



$$u(t) \leftrightarrow \frac{1}{s} \quad \Re(s) > 0$$

ramp



$$tu(t) \leftrightarrow \frac{1}{s^2} \quad \Re(s) > 0$$

exponential



$$e^{at}u(t) \leftrightarrow \frac{1}{s-a} \quad \Re(s) > a$$

2.) a $F(s) = \frac{6}{s+4}$

$$\mathcal{L}^{-1}\left\{\frac{6}{s+4}\right\} = 6e^{-4t}u(t)$$

b. $F(s) = \frac{10}{s(s+5)}$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{s+5}\right\}$$

$$f(t) = 2u(t) - 2e^{-5t}u(t)$$

c.

$$F(s) = \frac{s}{s+2} = 1 - \frac{2}{s+2}$$

$$f(t) = \mathcal{L}^{-1}\{1\} + \mathcal{L}^{-1}\left\{\frac{-2}{s+2}\right\}$$

$$= \delta(t) - 2e^{-2t}u(t)$$